PAPER

Locating and Mitigating Factual Hallucination Circuits in Small Language Models via Activation Patching

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SUMMARY Small language models (SLMs) are increasingly deployed in resource-constrained systems, yet they remain prone to factual hallucination, particularly when a misleading context conflicts with parametric knowledge (a context–memory conflict). We present a systematic activation-patching study of this conflict in three sub-2B models (Gemma-2-2B, Llama-3.2-1B, Qwen2.5-1.5B). Using a CounterFact-based benchmark of 2,000 facts that each model demonstrably knows, we first localize the components causally involved in answering under misleading context: restoration effects concentrate in the final quarter of the network across all three architectures (Spearman rank correlation of layer profiles 0.81–1.00), with a small set of late-layer attention heads dominating. Bidirectional patching reveals that these heads are *superposed*—each both conveys the misleading context and carries the memorized answer—and separates dual-role workhorses from conflict-specialist heads that are causally inert without a conflict. Knocking out the single strongest specialist head reduces misleading-context hallucination by 30–50% relative across models while *increasing* factual accuracy by up to 5.3 points, with MMLU retention indistinguishable from baseline; in contrast, norm-based head selection *increases* hallucination. We release quantified dose–response trade-offs between hallucination reduction and capability, and a selection protocol that guards against circular analysis.

key words: *language models, hallucination, mechanistic interpretability, activation patching, attention heads, context–memory conflict*

1. Introduction

Language models of 1–2B parameters now power on-device assistants, retrieval-augmented pipelines, and document-processing systems where latency and memory budgets preclude larger models. In such deployments the model must frequently integrate an external context with its parametric knowledge. When the context asserts a false fact—whether through a retrieval error, an adversarial document, or a mistaken user—the model faces a *context–memory (CM) conflict* [5], and resolving it incorrectly produces a factual hallucination.

Why a mechanistic account rather than a purely behavioral mitigation? Deployed systems need interventions that are cheap, targeted, and auditable: knowing *which* components carry the conflict lets a system builder trade hallucination against capability with measured confidence, instead of retraining, prompt-tuning blindly, or trusting an opaque detector. The account is more valuable still if the responsible components—and hence the mitigation recipe—transfer across architectures, which is why replication is a first-class concern in this work rather than an afterthought.

A growing body of work studies hallucination mechanisms, including knowledge-deficit failures [2] and probes/directions for truthfulness [3], [4]. Head-level analyses of CM conflict, however, have appeared mainly in multimodal settings [14], and recent work on knowledge-conflict interventions reports that influential attention heads play *superposed* dual roles [13]. What has been missing for sub-2B SLMs is a systematic causal account: which components, located how, trade hallucination against general capability when intervened upon, and whether such findings replicate across architectures.

This paper provides that account with a design that separates localization from validation: heads are selected only on a development split of relations, and all intervention results are reported on held-out relations. Our contributions are:

- **Localization.** A coarse-to-fine activation-patching map of the CM-conflict circuit in Gemma-2-2B (and its replication in two other architectures): restoration effects concentrate in late layers, and the attention/MLP decomposition shows the conflict is attention-dominated, unlike the MLP-heavy picture of classical fact recall.
- **Characterization.** Bidirectional patching quantifies superposition per head and separates dual-role heads from conflict-specialists that are causally inert in normal operation.
- **Quantified intervention trade-offs.** Dose–response curves for knockout and mean-ablation, with fairness baselines: patching-guided selection reduces hallucination significantly (McNemar $p < 10^{-6}$ even for a single head), whereas activation-norm selection *increases* it; capability costs are dissociated (MMLU retention ≈ 1.00; perplexity +11–20%).

2. Related Work

Hallucination and knowledge conflict. Surveys taxonomize hallucination in NLG [6] and knowledge conflicts in LLMs [5]. Behavioral studies quantify when context overrides memory [7]; Chuang et al. detect contextual hallucinations from attention maps [8]. Yu et al. trace non-factual hallucination to knowledge deficits in MLPs and selection failures in attention [2]; our corruption regime instead targets *knows-but-overridden* failures. Mechanistic work has

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also examined conflicts inside parametric memory itself [18] and cautioned that heads appearing to implement factual recall may partly implement generic copying [19]—a confound our neutral-control design quantifies directly.

Activation patching and circuits. Causal mediation introduced the method [9]; ROME localized factual associations with causal tracing [1]; best practices are systematized in [10]. Head-level analyses include IOI [11] and copy suppression [12]. Closest to us, JuICE reports head superposition in knowledge conflicts [13]; we test its claim at head granularity in SLMs under a counterfactual-distractor regime and exploit the resulting taxonomy for intervention.

Inference-time intervention. ITI steers along probe-identified truthful directions [3]; truth directions are linearly represented [4]; hallucination can also be detected from sampling uncertainty [17]. Our interventions are *subtractive* (head knockout/mean-ablation) and selected by causal patching rather than probes, with explicit capability trade-off curves.

Attention sinks and knowledge editing. Autoregressive models allocate disproportionate attention to initial tokens ("attention sinks"), which stabilize generation but are usually treated as functionally inert no-ops [15]. Our findings add a twist: the conflict-specialist heads that dominate the CM-conflict circuit are BOS-sink heads that are causally *active* under misleading context (Sect. 4.2). Separately, knowledge editing locates and rewrites stored facts [1], [16]; we instead locate components involved in *context override* and ablate them at inference time, leaving parameters untouched.

3. Experimental Setup

Fig. 1 summarizes the protocol.

Models. Primary: Gemma-2-2B (26 layers, 8 heads). Cross-model: Llama-3.2-1B (16L/32H), Qwen2.5-1.5B (28L/12H). All experiments use bfloat16 on a single 24GB GPU; a parity check between our hooking framework and reference HuggingFace logits passed in float32 (mean KL 1.3×10^{-5}, argmax agreement 100%), bounding numerical noise two to three orders below measured effects.

Data. From CounterFact [1] we retain facts each model demonstrably knows (greedy completion equals the true target and the first-token logit gap over the counterfactual distractor ≥ 0.5), sampling 2,000 facts (16.7% survival of 12,000 scored) split 807/1,193 dev/held-out *by relation*, so the held-out split shares no relation with the split used for any selection. A CM-conflict suite asserts the distractor in three templates; a neutral control replaces it with a non-answer token.

Metrics and definitions. Let $z_t(x)$ and $z_d(x)$ denote the logits, at the answer position of input x, of the first tokens of the true target and the distractor, respectively. The *logit difference* is $\mathrm{LD}(x) = z_t(x) - z_d(x)$. Writing x_{corr} for the corrupted input and x_{patched} for the corrupted input with a clean activation substituted at some component, the normalized *restoration effect* is

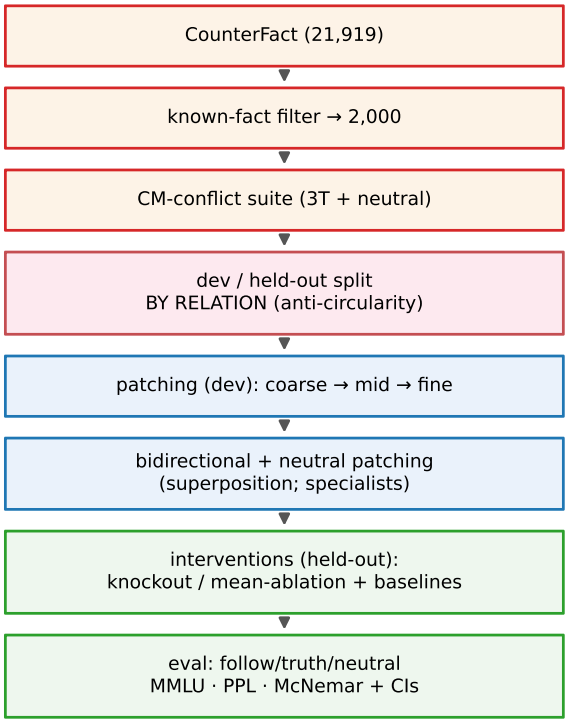


Fig. 1 Protocol overview. Data construction (top), causal localization on the dev split (middle), and interventions evaluated exclusively on held-out relations (bottom).

$$E = \frac{\mathrm{LD}(x_{\mathrm{patched}}) - \mathrm{LD}(x_{\mathrm{corr}})}{\mathrm{LD}(x_{\mathrm{clean}}) - \mathrm{LD}(x_{\mathrm{corr}})}, \tag{1}$$

so $E = 0$ under no restoration and $E = 1$ under full restoration of the clean margin. For each influential head we compute three directed effects via Eq. (1): the *restore effect* A (clean value patched into the misleading run), the *context-push effect* B (misleading value patched into the clean run), and the *neutral effect* N (clean value patched into a neutral-context run). The *superposition index* $S = |A-B|/(|A|+|B|)$ equals 1 for a head that pushes one side only and 0 for a fully dual-role head. Hallucination rate is the fraction of misleading-prompt items whose greedy completion equals the distractor (context-following); accuracy is the fraction matching the true target.

Protocol. Stage 1 (coarse): patch residual-stream block outputs layer-by-layer under two corruptions—ROME-style embedding noise, and the misleading context itself—measuring the normalized restoration of the true-vs-distractor logit difference. Stage 2 decomposes attention vs. MLP outputs on top layers; Stage 3 patches per-head z-vectors. Stage 4: bidirectional and neutral-context patching characterize each top head (A, B, N; S as above). Stage 5 intervenes by zeroing or mean-ablating selected heads and evaluates hallucination rate, accuracy, and neutral accuracy on held-out items; MMLU (2,000-question fixed subset) and WikiText-2 perplexity (1024-token windows) measure capability under the same hooks. Fairness baselines ablate

Table 1 Behavioral landscape on known facts (Gemma-2-2B, n=2,000; misleading suite per template).

condition	context-following	truth
clean (no context)	—	100%
misleading T0	24.2%	66.4%
misleading T1 (adjacent distractor)	98.4%	1.1%
misleading T2	19.2%	71.0%
neutral control	1.2%†	87.4%

†fraction copying the neutral token.

randomly selected heads (5 seeds) and top-|z|-norm heads. Statistics: bootstrap 95% CIs; McNemar tests against the no-intervention condition.

4. Results

4.1 Behavioral landscape and localization

On known facts, the bare-prompt accuracy of Gemma-2-2B is 100%. Under misleading context the competition is nearly balanced overall (47.3% context-following vs. 46.2% truth; CI 46.0–48.5) but strongly template-dependent (24.2%/98.4%/19.2% following across templates), matching persuasion-strength variation reported behaviorally [7]; the neutral control stays at 87.4% truth with 1.2% copying of the neutral token, bounding pure token-copy effects (Table 1). Template 1, which places the distractor immediately before the repeated question, drives following to 98.4%—an upper bound on recency/copy contributions that the neutral control and bidirectional analyses below separate from belief override. Reference benchmarks: MMLU 49.4%, TruthfulQA MC1/MC2 24.5%/36.3%, WikiText-2 PPL 65.3.

Fig. 2 shows the coarse maps. Restoration under the misleading regime concentrates in the final layers, peaking at the last layer (0.333); the noise regime peaks earlier (layer 21, 0.227). Table 2 decomposes the top blocks: the misleading-conflict signal is attention-dominated (L22 attention 0.121 vs. ≤0.01 for every MLP block), whereas pure recall restoration under noise mixes MLP and attention almost evenly. This contrast is itself informative: the classical ROME-style picture of factual recall as MLP-stored knowledge [1] does not transfer to the conflict regime, where what an intervention can restore flows through attention. Head-level patching then identifies a compact set: {L22.H0 (0.26), L25.H4 (0.24), L20.H3 (0.15), L25.H6 (0.12)} under noise, and {L25.H4 (0.08), L22.H4 (0.07), L22.H0, L20.H3} under the misleading context—note the partial overlap: three of four noise-top heads reappear in the conflict ranking, and L22.H4 emerges only under conflict.

4.2 Superposition and conflict specialists

Bidirectional patching (Fig. 4) shows every influential head conveys the misleading context more strongly than it restores the memory ($B > A$; superposition indices 0.21–0.62), confirming head-level superposition [13] in this regime. The

Table 2 Block-level decomposition (mean restoration effect, top layers).

block	noise regime	misleading regime
L22 attention	0.074	**0.121**
L25 attention	0.018	0.048
L21 MLP	**0.076**	≤0.01
L20 MLP	0.058	≤0.01

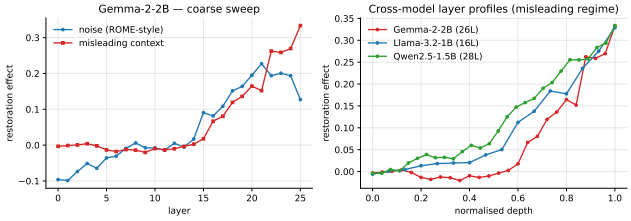


Fig. 2 Layer profiles of restoration effect. (a) Gemma-2-2B under both corruption regimes; (b) misleading-regime profiles across the three architectures at normalized depth. Late-layer concentration replicates (pairwise Spearman 0.81–1.00).

Table 3 Per-head characterization (Gemma-2-2B, top-10 by restore effect A; S = superposition index; N/A = neutral ratio).

head	A	B	N	S	N/A
L25.H4	0.081	0.134	0.062	0.24	0.76
L22.H4	0.071	0.143	−0.006	0.34	−0.09
L22.H0	0.048	0.095	0.002	0.33	0.05
L20.H3	0.038	0.096	−0.004	0.43	−0.10
L23.H7	0.016	0.038	−0.012	0.41	−0.72
L23.H4	0.012	0.022	−0.013	0.29	−1.02
L20.H7	0.010	0.042	−0.009	0.62	−0.91
L22.H3	0.010	0.030	−0.013	0.50	−1.33
L25.H2	0.010	0.024	0.004	0.42	0.38
L21.H1	0.009	0.019	0.001	0.34	0.06

neutral control separates two populations: L25.H4 retains 76% of its effect without a conflict (generic memory role), whereas L22.H4, L22.H0 and L20.H3 vanish ($N \approx 0$)—*conflict specialists* that are causally inert in normal operation. Fig. 3 adds a mechanistic signature: on the misleading prompt, the specialists concentrate their attention on the beginning-of-sequence token (0.92–1.00)—attention-sink heads that contribute to the conflict through the value pathway rather than by reading the distractor directly—whereas the dual-role heads L25.H4/L25.H6 attend to the template tokens asserting the distractor (0.67–0.78). Table 3 reports the full per-head characterization for the ten highest-restore heads. Two structure-level observations follow. First, $B > A$ for *every* head—no influential head is a pure "memory carrier"; all participate in conveying the context. Second, the specialist cluster (L22.H4, L22.H0, L20.H3) combines $N \approx 0$ with mid-range superposition indices (0.33–0.43), whereas strongly superposed heads with large $|N/A|$ (L20.H7, L22.H3) sit lower in the restore ranking: generic-role heads tend to be both superposed and less useful as intervention targets.

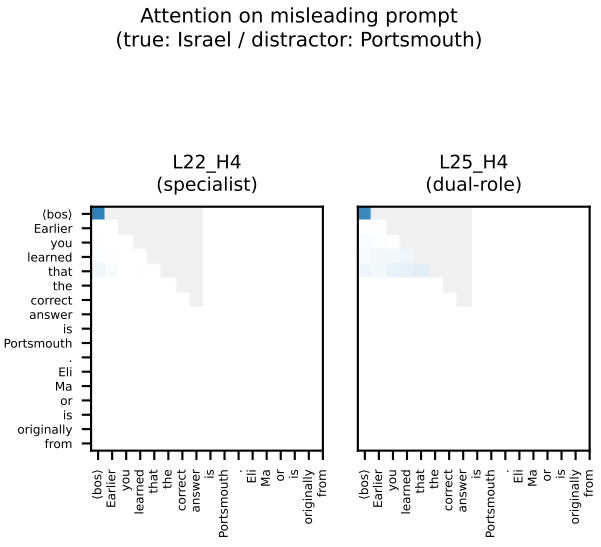


Fig. 3 Attention patterns of the specialist head L22.H4 (left) and the dual-role head L25.H4 (right) on a representative misleading prompt ("Eli Maor is originally from"; true: Israel, distractor: Portsmouth). The specialist attends almost exclusively to ⟨bos⟩; the dual-role head spreads attention over the template tokens asserting the distractor.

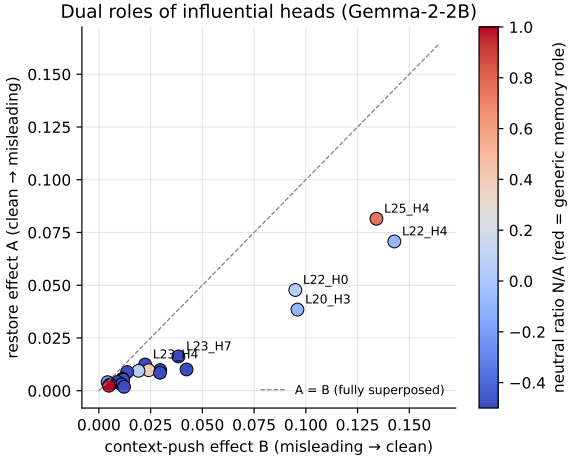


Fig. 4 Dual roles of the top-20 heads: context-push effect B vs. restore effect A, colored by neutral ratio N/A. Points near the diagonal are superposed; L22.H4 combines the strongest context conveyance with zero neutral role.

4.3 Interventions and quantified trade-offs

Table 4 and Fig. 5 report the held-out interventions. Knocking out the dev-ranked top-k heads reduces context-following monotonically (19.3%→1.6% at k=20) but at matching accuracy cost—a blunt instrument. The E4 taxonomy, however, predicts that removing a conflict specialist should suppress the misleading pull without collateral damage. This is borne out: knocking out the single head L22.H4 reduces hallucination to 12.6% (McNemar b=81, c=1, $p \approx 0$) while *increasing* accuracy to 79.7% (+5.3 points) and leaving neutral accuracy at 92.0% (baseline 93.5%). By contrast, ablating the five highest-norm heads *increases* hallucination to 33.0% (b=9, c=173): they are the dual-role workhorses, and their removal tips the final balance toward the context. Random-head ablation produces erratic, non-significant effects.

Table 4 Held-out interventions (Gemma-2-2B, misleading template T0). Δacc. is vs. the none condition.

config	follow↓	truth↑	neutral
none	19.3%	74.4%	93.5%
top-1	15.8%	66.8%	82.6%
top-2	11.7%	69.9%	80.1%
top-5	8.5%	44.7%	50.9%
top-20	1.6%	19.4%	25.9%
spec1 (L22.H4 only)	**12.6%**	**79.7%**	**92.0%**
spec3	10.9%	63.3%	68.1%
mean-abl. top-5	8.8%	46.4%	53.8%
topnorm-5	33.0%	59.1%	85.9%
random-5 (mean)	18.9%	—	—

Table 5 Capability under intervention (fixed 2,000-question MMLU subset; PPL over 1024-token WikiText-2 windows).

config	MMLU	retention	PPL ratio
none	50.0%	1.00	1.00
top-2	49.7%	0.99	1.11
top-5	50.5%	1.01	1.20
spec1 (L22.H4 only)	**49.9%**	**1.00**	**1.05**
spec3	50.4%	1.01	1.16
mean-abl. top-5	50.0%	1.00	1.13

Capability costs are dissociated (Table 5): MMLU retention is statistically indistinguishable from baseline for every tested intervention (0.99–1.01), whereas perplexity rises 5–20%. The single-head specialist intervention is the cheapest of all: a 35% relative hallucination reduction—with an accuracy *gain*—at MMLU retention 0.998 and a perplexity cost of only 5%.

Template robustness. Re-evaluating the none, L22.H4, and top-5 conditions on the two remaining held-out templates tests whether the specialist's effect depends on how the context asserts the distractor. On T2 (user-hint template) L22.H4 again reduces following (16.6%→11.1%) while raising accuracy by 3.3 points, replicating T0. On T1—the extreme adjacent-distractor template whose baseline behavior is nearly pure context-following (98.4%) and which the neutral-control analysis attributes largely to recency copying—L22.H4 still cuts following by 9 points and increases accuracy tenfold (0.8%→8.7%), while top-5 removes far more (58.3%): consistent with T1 additionally recruiting a copy pathway that a single specialist head does not fully control. The intervention thus generalizes across persuasion formats, with residual copy-mediated following as the honest boundary of the single-head approach.

4.4 Cross-model replication

Replicating the pipeline on Llama-3.2-1B and Qwen2.5-1.5B (600 known facts each; 300/150 facts for coarse/fine

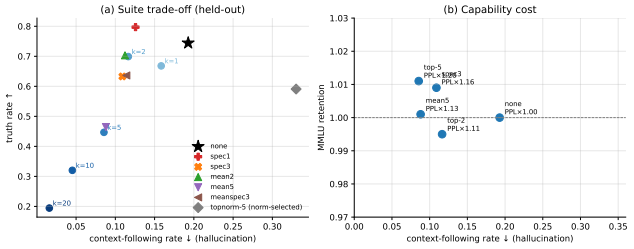


Fig. 5 Trade-offs. (a) suite space: hallucination vs. truth; the specialist point (red P) dominates the knockout frontier. (b) capability space: hallucination vs. MMLU retention (PPL ratios annotated).

Table 6 Cross-model replication summary (misleading regime, held-out).

	Gemma-2-2B	Llama	Qwen
layers	26	16	28
top coarse layers	20–25	13–15	22–27
facts used	2,000	600	600
follow (none)	19.3%	6.6%	14.9%
follow (top-1)	12.6%*	4.6%	7.5%
Δ truth	+5.3*	+0.5	+0.5
MMLU retention	.99–1.01	1.01	1.00

*Gemma-2-2B uses the E4-selected specialist (L22.H4); the other models use their own dev-ranking top-1. Gemma-2-2B retention spans the tested interventions (top-2/5, spec3, mean-abl.).

sweeps) reproduces every headline finding (Table 6). First, late-layer concentration holds in all three architectures: the top-5 coarse layers are 20–25 of 26 (Gemma-2-2B), 13–15 of 16 (Llama), and 22–27 of 28 (Qwen), and the normalized-depth profiles rank-correlate at 0.81 (Gemma-2-2B vs. each other model) and 1.00 (Llama vs. Qwen). Second, the pipeline's behavioral entry point replicates with model-dependent baseline levels (following 6.6% and 14.9%—weaker and stronger than Gemma-2-2B's 19.3% on the held-out split), reflecting different context persuadability. Third, and most importantly, the single-head knockout from each model's own dev ranking reduces hallucination by 30% (Llama) and 50% (Qwen) with accuracy changes of +0.5/+0.5 points and MMLU retention 1.01/1.00—the same qualitative signature as Gemma-2-2B's specialist result, obtained without any per-model specialist analysis (E4 was run only for Gemma-2-2B), suggesting the effect is a property of the patching-guided selection rather than of one architecture. Per-head restoration effects are smaller in the 1–1.5B models (0.04–0.06 vs. 0.08), consistent with a more distributed conflict circuit at smaller scale, which makes the preserved single-head intervention effect all the more notable.

5. Discussion

Mechanism. The CM conflict is resolved late and attentionally: answer-position representations in the final layers carry most of what an intervention can restore, and the influential heads are conduits of the misleading context at least as much as carriers of memory ($B > A$ throughout). "Restoration" by patching therefore partly works by displacing a head's context conveyance rather than injecting memory—the neutral-control analysis makes this distinction measurable, addressing copy-vs-recall confounds.

Attention-sink specialists. The most surprising mechanistic finding is the identity of the conflict specialists: heads whose attention is almost entirely absorbed by the beginning-of-sequence token. In the streaming-efficiency literature such sinks are treated as stabilizing but semantically inert no-ops that can be cached and forgotten [15]. Our results show a sink head can be causally central to a high-level behavior: L22.H4 conveys the strongest misleading-context push (B=0.143), does nothing without a conflict ($N \approx 0$), and its removal improves factual accuracy while leaving perplexity and MMLU essentially intact. We conjecture that sink-focused heads provide a low-semantic-content channel whose residual contribution modulates how strongly downstream layers weigh contextual assertions—a hypothesis path-patching follow-ups could test directly. This also reframes the folk assumption that "a head that only attends to BOS is a no-op": sink heads should be included in behavioral circuit searches rather than filtered out.

Design implication. The dissociation between MMLU (intact) and perplexity (degraded) suggests these heads implement contextual integration that matters for language modeling loss but not for knowledge-task accuracy; removing the right one is nearly free for deployment QA settings while reducing hallucination. The contrast with norm-based selection carries a practical warning: cheap proxies for "importance" can select exactly the wrong components and *worsen* the targeted behavior.

Limitations and future work. (i) English-only, single-decoding (greedy), completion-style evaluation; (ii) hallucination is operationalized as verifiable counterfactual override, not open-ended factuality; (iii) head selection is causal but the circuit description is coarse—path-level analysis (direct vs. mediated effects) is the natural next step; (iv) Gemma-2-2B is not the newest generation—chosen for tooling maturity and comparability with the interpretability literature; newer families (Gemma 3/4) and instruction-tuned variants are future work; (v) capability is proxied by MMLU/PPL subsets; (vi) the conflict-specialist analysis (E4) was run only for Gemma-2-2B; extending the specialist taxonomy to the replication models, and testing whether entity popularity modulates circuit reliance, remain open.

6. Conclusion

We mapped, validated, and exploited the CM-conflict circuit of SLMs: late-layer attention heads with measurable superposition, among which conflict specialists can be switched off individually—hallucination falls by 30–50% relative, accuracy rises, and knowledge-task capability is retained—while norm-based selection backfires. All selection was performed on dev relations with results reported on held-out relations, and the protocol replicated across three architectures. We release code, data, and per-item records for

reproduction on a single consumer GPU.

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